Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A wave has frequency 4 Hz and wavelength 6 m. Find its speed.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: λ = v/f; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

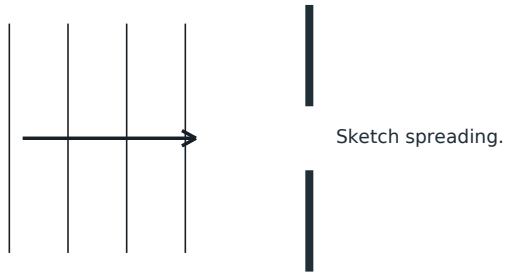
Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

Marzano check - circle: ① ② ③

03 Look and explain

Sketch wavefronts after a narrow opening about one wavelength wide. Compare with a much wider opening.



Marzano check - circle: ① ② ③

04 Build the solution

Opening A is 1 m wide, opening B is 10 m; wavelength is 1 m. Which shows more noticeable spreading relative to opening width? Explain.

Marzano check - circle: ① ② ③

05 Use the idea

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Marzano check - circle: 1 2 3

06 Choose a method

At the same kelvin temperature, sample B contains 4 times as many particles of the same ideal gas as A. Compare average energy per particle and total translational energy.

Marzano check - circle: 1 2 3

07 Exit check

For the same opening, would a longer wavelength generally spread more or less?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Keep opening width fixed and halve wavelength. Predict the change in relative spreading and draw before/after patterns.

Marzano check - circle: 1 2 3